Southeast University

Department of Computer Science and Engineering (CSE)

School of Sciences and Engineering

Semester: (Summer, Year: 2026)

LAB REPORT NO: **03**

Course Title: Operating Systems Lab

Course Code: CSE362.2

Batch: 65

Lab Experiment Name: Implementing basic I/O operations, arithmetic operations, and conditional operations.

Student Details

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Submission Date : **24-07-26**

Lab Report Status

Marks: ………………………………… **Signature:.....................**
Comments:... **Date:..............................**

Introduction:

Bash scripting is an essential tool in Unix-like operating systems that allows users to automate tasks and interact with the operating system via the command line. This lab experiment focuses on understanding the core concepts of shell scripting using Bash. It covers reading user input and printing output, performing standard arithmetic calculations, and utilizing conditional statements for both numerical comparisons and string evaluations. Mastering these foundational scripting concepts provides a strong basis for system administration and process control in operating systems.

Objectives:

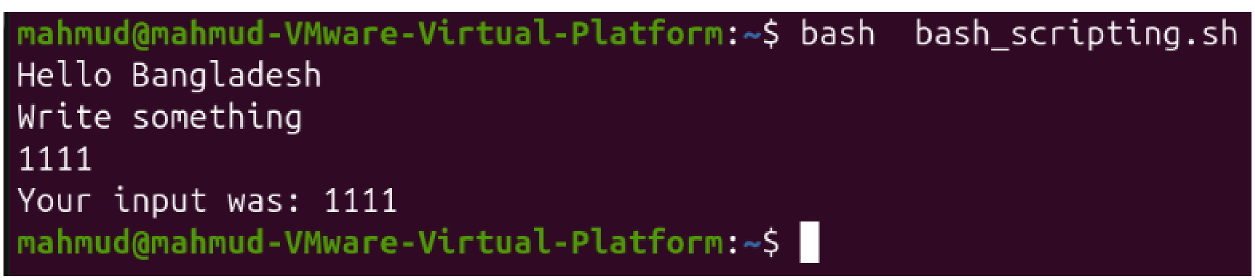
- Master Basic Input/Output (I/O) Operations: Learn how to prompt users for input using the read command and display formatted output using the echo command in Bash.
- Implement Arithmetic Calculations: Perform basic mathematical operations using shell arithmetic expansion.
- Apply Conditional Logic: Implement conditional structures to make decisions based on numerical equality, check for even/odd values, and perform string matching.
- Gain Hands-On Terminal Experience: Write, save, and execute shell scripts inside a Linux environment.

Implementation 1: Basic I/O operation

Bash code:

```
#!/bin/bash
echo "Hello Bangladesh"
echo "Write something"
read a
echo "Your input was: $a"
```

Screenshot:



Implementation 2: Arithmetic operation

Bash code:

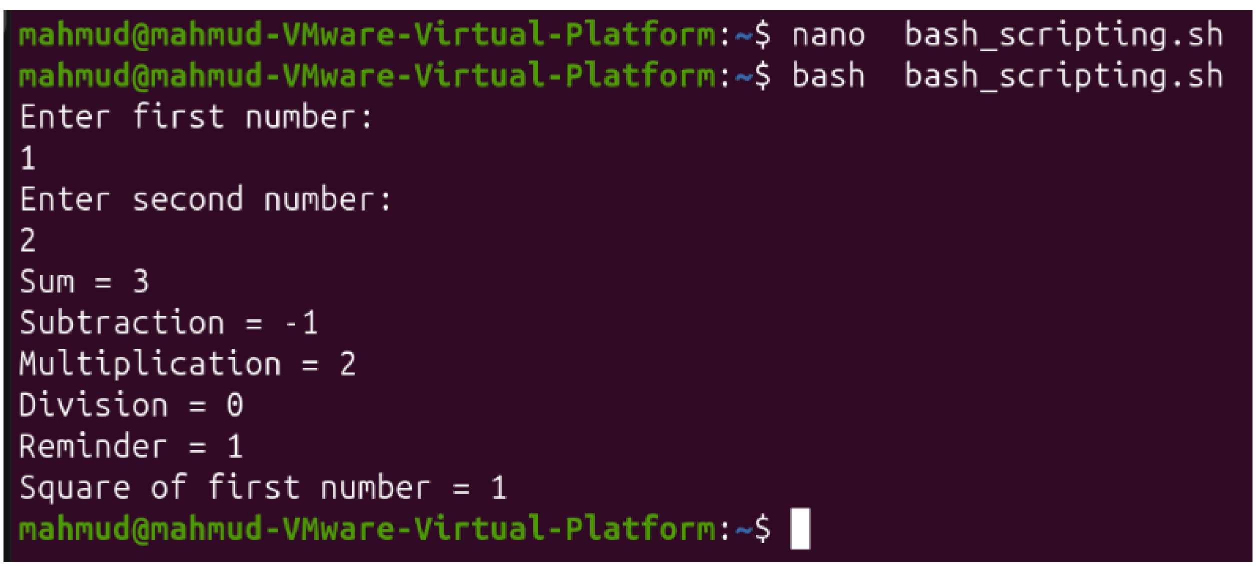
```
#!/bin/bash
echo "Enter first number:"
```

```
read a
echo "Enter second number:"
read b

sum=$((a+b))
sub=$((a-b))
mul=$((a*b))
div=$((a/b))
rem=$((a%b))
sq=$((a**2))

echo "Sum = $sum"
echo "Subtraction = $sub"
echo "Multiplication = $mul"
echo "Division = $div"
echo "Reminder = $rem"
echo "Square of first number = $sq"
```

Screenshot:



Implementation 3: Basic I/O operation

Bash code:

```
#!/bin/bash
echo "Enter first number:"
read a
echo "Enter second number:"
read b
```

```
if [[ $a -eq $b ]]
then
   echo "Numbers are equal"
else
   echo "Numbers are not equal"
fi

echo "Enter a number to check Even or Odd:"
read a

div=$((a % 2))

if [[ $div -eq 0 ]]
then
   echo "Even"
else
   echo "Odd"
fi
```

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ nano  bash_scripting.sh
mahmud@mahmud-VMware-Virtual-Platform:~$ bash  bash_scripting.sh
Enter first number:
5
Enter second number:
6
Numbers are not equal
Enter a number to check Even or Odd:
6
Even
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Implementation 4: Basic I/O operation

Bash code:

```
#!/bin/bash

echo "Enter a string:"
read str
```

```
if [[ $str == "CSE" ]]
then
   echo "The string is CSE"
else
   echo "The string is not CSE"
fi
```

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ bash  bash_scripting.sh
Enter a string:
Mahmud 799
The string is not CSE
mahmud@mahmud-VMware-Virtual-Platform:~$ bash  bash_scripting.sh
Enter a string:
CSE
The string is CSE
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Conclusion:

In this experiment, basic Bash scripting concepts were successfully implemented and executed in a Linux terminal environment. Hands-on experience was gained in handling user input and output, performing arithmetic calculations, and controlling execution flow using conditional statements. Understanding these fundamental constructs such as checking for number equality, evaluating odd/even conditions, and comparing strings provides a strong foundation for developing more complex scripts to automate operating system tasks efficiently.